

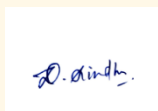
SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCHEMA ANALYSIS AND VIVA VOCE

Course Code:	DSA0502	Course Title:	Query Processing for Data Science With Real Time Applications
CO Assessed:	CO2	Assessment:	Tool 2 – Schema Analysis and Viva Voce
Weightage:		Total Marks:	35 Marks
CO2 Statement: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BL6)			
Student Name:	D. Sindhu	Reg. No.:	192424197
Section / Batch:	2024	Date of Submission:	02.08.2026
Faculty In-charge:	Dr. B.SHYAMALA GOWRI	Project Mode:	Individual Submission

Code of Conduct

I D. Sindhu (192424197) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature: _____

Identification of Entities and Attributes (5 Marks)	Understanding of Relationships (5 Marks)	Analysis of Keys and Constraints (5 Marks)	Detection of Design Issues (10 Marks)	Viva Voce (5 Marks)	Total (35 Marks)

1. University Course Registration System

A university database contains the following tables:

- Student(StudentID, Name, DepartmentID)
- Department(DepartmentID, DepartmentName)
- Course(CourseID, CourseName, FacultyID)
- Faculty(FacultyID, FacultyName)
- Enrollment(StudentID, CourseID, Semester)

A student can enroll in multiple courses, and each course can have multiple students.

Viva Questions

Q1. Why is the Enrollment table necessary instead of storing CourseID directly in the Student table?

Q2. If the university wants to track grades for each enrolled course, how would you modify the schema?

2. Hospital Management System

A hospital database contains:

- Patient(PatientID, Name, Age)
- Doctor(DoctorID, DoctorName, Specialization)
- Appointment(AppointmentID, PatientID, DoctorID, Date)
- Prescription(PrescriptionID, AppointmentID, Medicine)

Multiple patients can consult the same doctor.

Viva Questions

Q1. What problems might occur if prescription details are stored directly in the Appointment table?

Q2. How would the schema change if a prescription contains multiple medicines?

3. Online Shopping System

Database tables include:

- Customer(CustomerID, Name)
- Product(ProductID, ProductName, Price)
- Order(OrderID, CustomerID, OrderDate)
- OrderDetails(OrderID, ProductID, Quantity)

Each order may contain multiple products.

Viva Questions

Q1. Why is the OrderDetails table required in this schema?

Q2. What would happen if ProductID were stored directly in the Order table?

4. Library Management System

Database tables:

- Member(MemberID, Name)
- Book(BookID, Title, Author)
- Issue(MemberID, BookID, IssueDate, ReturnDate)

Members can borrow multiple books.

Viva Questions

Q1. Explain the relationship between Member and Book through the Issue table.

Q2. How would you modify the schema to track overdue fines?

5. Banking System

Tables:

- Customer(CustomerID, Name)
- Account(AccountNo, CustomerID, Balance)
- Transaction(TransactionID, AccountNo, Amount, Type)

One customer can have multiple accounts.

Viva Questions

Q1. Why should transactions be stored separately from account details?

Q2. How would you redesign the schema to support joint accounts?

6. Employee Payroll System

Tables:

- Employee(EmployeeID, Name, DepartmentID)
- Department(DepartmentID, DepartmentName)
- Payroll(PayrollID, EmployeeID, Salary, Month)

Employees belong to departments and receive monthly salaries.

Viva Questions

Q1. Why is Department information separated into a different table?

Q2. What redundancy issues would arise if department details were stored in Employee records?

7. Hotel Reservation System

Tables:

- Customer(CustomerID, Name)
- Room(RoomID, RoomType)
- Reservation(ReservationID, CustomerID, RoomID, CheckIn, CheckOut)

Rooms can be booked by different customers at different times.

Viva Questions

Q1. Why should reservation details be maintained separately from room details?

Q2. How would you prevent double booking of rooms using the schema?

8. Food Delivery Application

Tables:

- Customer(CustomerID, Name)
- Restaurant(RestaurantID, Name)
- Order(OrderID, CustomerID, RestaurantID)
- DeliveryAgent(AgentID, Name)

- Delivery(OrderID, AgentID)

Orders are delivered by delivery agents.

Viva Questions

Q1. Why is Delivery maintained as a separate table?

Q2. How would the schema change if multiple delivery agents could handle a single order?

9. Learning Management System (LMS)

Tables:

- Student(StudentID, Name)
- Course(CourseID, CourseName)
- Enrollment(StudentID, CourseID)
- Assignment(AssignmentID, CourseID)

Students can enroll in many courses.

Viva Questions

Q1. Why is Enrollment considered a bridge table?

Q2. How would you modify the schema to record assignment grades?

10. Bus Ticket Reservation System

Tables:

- Passenger(PassengerID, Name)
- Bus(BusID, Route)
- Booking(BookingID, PassengerID, BusID, TravelDate)

Passengers can make multiple bookings.

Viva Questions

Q1. Why should booking details be stored separately from passenger information?

Q2. If a booking contains multiple passengers, how would you redesign the schema?

1. University Course Registration System

Q1. Why is the Enrollment table necessary instead of storing CourseID directly in the Student table?

Answer:

- A student can enroll in many courses.
- A course can have many students.
- This is a **many-to-many relationship**, so the **Enrollment** table is needed.

Q2. If the university wants to track grades for each enrolled course, how would you modify the schema?

Answer:

Add a **Grade** attribute to the Enrollment table.

Enrollment(StudentID, CourseID, Semester, Grade)

2. Hospital Management System

Q1. What problems might occur if prescription details are stored directly in the Appointment table?

Answer:

- One appointment may have many medicines.
- It causes data duplication.
- It is difficult to update prescription details.

Q2. How would the schema change if a prescription contains multiple medicines?

Answer:

Create a new table:

PrescriptionMedicine(PrescriptionID, MedicineName, Dosage)

3. Online Shopping System

Q1. Why is the OrderDetails table required in this schema?

Answer:

- One order can contain many products.
- One product can appear in many orders.
- OrderDetails stores product quantity and links orders with products.

Q2. What would happen if ProductID were stored directly in the Order table?

Answer:

- Only one product could be stored per order.
 - Multiple-product orders would not be possible.
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4. Library Management System

Q1. Explain the relationship between Member and Book through the Issue table.

Answer:

- A member can borrow many books.
- A book can be borrowed by different members over time.
- **Issue** acts as the bridge table.

Q2. How would you modify the schema to track overdue fines?

Answer:

Add a **FineAmount** column.

Issue(MemberID, BookID, IssueDate, ReturnDate, FineAmount)

5. Banking System

Q1. Why should transactions be stored separately from account details?

Answer:

- One account has many transactions.
- Separate storage keeps transaction history.
- It avoids repeated account information.

Q2. How would you redesign the schema to support joint accounts?

Answer:

Create a bridge table:

CustomerAccount(CustomerID, AccountNo)

This allows multiple customers to share one account.

6. Employee Payroll System

Q1. Why is Department information separated into a different table?

Answer:

- Many employees belong to the same department.
- It reduces duplicate department data.
- It improves data consistency.

Q2. What redundancy issues would arise if department details were stored in Employee records?

Answer:

- Department names would be repeated.
 - Updating department information becomes difficult.
 - It causes data inconsistency.
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7. Hotel Reservation System

Q1. Why should reservation details be maintained separately from room details?

Answer:

- One room can have many reservations at different times.
- Reservation stores booking information.
- Room stores only room details.

Q2. How would you prevent double booking of rooms using the schema?

Answer:

- Check whether the room is already reserved for the selected dates before inserting a new reservation.
 - Do not allow overlapping reservations.
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8. Food Delivery Application

Q1. Why is Delivery maintained as a separate table?

Answer:

- Delivery information is different from order information.
- It connects orders with delivery agents.
- It avoids repeating agent details.

Q2. How would the schema change if multiple delivery agents could handle a single order?

Answer:

Use the Delivery table as:

Delivery(OrderID, AgentID, AssignedTime)

Each order can have multiple agent records.

9. Learning Management System (LMS)

Q1. Why is Enrollment considered a bridge table?

Answer:

- It connects students and courses.
- It represents a many-to-many relationship.

Q2. How would you modify the schema to record assignment grades?

Answer:

Create a new table:

Submission(StudentID, AssignmentID, Grade)

10. Bus Ticket Reservation System

Q1. Why should booking details be stored separately from passenger information?

Answer:

- A passenger can make many bookings.
- Booking stores travel details.
- Passenger stores personal information.

Q2. If a booking contains multiple passengers, how would you redesign the schema?

Answer:

Create a bridge table:

BookingPassenger(BookingID, PassengerID)

This allows one booking to include multiple passengers.
